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```
import java.util.*;
```

```
class Customer {
```

```
    private String name;
```

```
    private Order orderType;
```

```
    public static double grandTotal = 0.0;
```

```
    public Customer (String name, Order orderType) {
```

```
        this.name = name;
```

```
        this.orderType = orderType;
```

```
    }
```

```
    public void placeOrder (Food food) {
```

```
        System.out.println ( name + "is placing order" );
```

```
        orderType.orderFood (food);
```

```
    }
```


}

interface Order {

void orderFood (Food food);

}

class OfflineOrder implements Order {

public final double SERVICE_CHARGE = 2.0;

@Override

public void orderFood (Food food) {

food.prepare();

double cost = food.getPrice() + SERVICE_CHARGE;

Customer.grandTotal += cost;

System.out.println (food.getName() + " ordered offline.");

System.out.printf ("Total = RM %.2f\n" , cost);

System.out.println();

}

}


```
class OnlineOrder implements Order {
```

```
    public final double SERVICE_CHARGE = 1.0;
```

```
    @Override
```

```
    public void orderFood (Food food) {
```

```
        food.prepare();
```

```
        double cost = food.getPrice() + SERVICE_CHARGE;
```

```
        Customer.grandTotal += cost;
```

```
        System.out.println ( food.getName() + " ordered offline." );
```

```
        System.out.printf ( "Total = RM %.2f\n" , cost );
```

```
        System.out.println();
```

```
    }
```

```
}
```

```
abstract class Food {
```

```
    private String name;
```

```
    private double price;
```

```
    public Food ( String name, double price){
```

```
        this.name = name;
```

```
        this.price = price;
```


}

```
public String getName () { return name; }
```

```
public double getPrice () { return price; }
```

```
public abstract void prepare ();
```

}

```
class Pizza extends Food {
```

```
    public Pizza () {
```

```
        super ("Hawaiian Chicken", 40.90);
```

```
    }
```

① Override

```
    public void prepare () {
```

```
        System.out.println("Preparing Hawaiian Chicken Pizza ...");
```

```
    }
```

```
}
```

```
class Burger extends Food {
```

```
    public Pizza () {
```

```
        super ("Double Cheeseburger", 18.35);
```

```
    }
```


@ Override

```
public void prepare () {
```

```
    System.out.println( "Preparing Double Cheeseburger ... " );
```

```
}
```

```
}
```

```
class FoodOrderingSystem {
```

```
    public static void addCustomer ( Vector < Customer > cList, Customer c ) {
```

```
        cList.add ( c );
```

```
}
```

```
    public static void takeOrders ( Vector < Customer > cList ) {
```

```
        for ( Customer c : cList ) {
```

```
            c.placeOrder ( new Pizza() );
```

```
            c.placeOrder ( new Burger() );
```

```
}
```

```
}
```

```
    public static void main ( String [] args ) {
```

```
        Vector < Customer > cList = new Vector < > ();
```

```
        addCustomer ( cList, new Customer ( "Harris", new onlineOrder() ) );
```

```
        addCustomer ( cList, new Customer ( "Aliya", new offlineOrder() ) );
```



```
takeOrders(cList);
```

```
System.out.printf("Grand Total Sales = RM%.2f\n", Customer.grandTotal);
```

```
}
```

```
}
```


